

Java Practical Programs – 10 Solutions

Simple, exam-friendly Java programs based on the practical list provided.

Quick Revision

Q	Practical	Main Concept
1	Reverse String	String, loop
2	Anagram	char array, sorting
3	Palindrome	Reverse + comparison
4	Character Count	Frequency array
5	2D Array Sum	Nested loops
6	Prefix Sum	Cumulative sum
7	Two Pointer	Left and right pointers
8	Fibonacci	Loop + variables
9	Factorial	Multiplication loop
10	Remove Duplicates	indexOf()

Q1. Reverse a String

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();
        String reverse = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            reverse = reverse + str.charAt(i);
        }

        System.out.println(reverse);

        sc.close();
    }
}
```

Q2. Check Whether Two Strings Are Anagrams

```
import java.util.Scanner;
import java.util.Arrays;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str1 = sc.nextLine();
        String str2 = sc.nextLine();

        str1 = str1.replaceAll("\\s", "").toLowerCase();
        str2 = str2.replaceAll("\\s", "").toLowerCase();

        char[] a = str1.toCharArray();
        char[] b = str2.toCharArray();

        Arrays.sort(a);
        Arrays.sort(b);

        if (Arrays.equals(a, b)) {
            System.out.println("Anagram");
        } else {
            System.out.println("Not Anagram");
        }

        sc.close();
    }
}
```

Q3. Check Whether String or Number Is Palindrome

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();
        String reverse = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            reverse = reverse + str.charAt(i);
        }

        if (str.equalsIgnoreCase(reverse)) {
            System.out.println("Palindrome");
        } else {
            System.out.println("Not Palindrome");
        }

        sc.close();
    }
}
```

Q4. Count Characters From a Given String

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();
        int[] count = new int[256];

        for (int i = 0; i < str.length(); i++) {
            count[str.charAt(i)]++;
        }

        for (int i = 0; i < 256; i++) {
            if (count[i] > 0) {
                System.out.println((char)i + " = " + count[i]);
            }
        }

        sc.close();
    }
}
```

Q5. 2D Array – Find Sum of Matrix

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();
        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];
        int sum = 0;

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                matrix[i][j] = sc.nextInt();
                sum = sum + matrix[i][j];
            }
        }

        System.out.println("Sum = " + sum);

        sc.close();
    }
}
```

Q6. Prefix Sum

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];
        int[] prefix = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        prefix[0] = arr[0];

        for (int i = 1; i < n; i++) {
            prefix[i] = prefix[i - 1] + arr[i];
        }

        for (int i = 0; i < n; i++) {
            System.out.print(prefix[i] + " ");
        }

        sc.close();
    }
}
```

Q7. Two Pointer – Left and Right

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int left = 0;
        int right = n - 1;

        while (left < right) {
            int temp = arr[left];
            arr[left] = arr[right];
            arr[right] = temp;

            left++;
            right--;
        }

        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }

        sc.close();
    }
}
```

Q8. Fibonacci Series

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int first = 0;
        int second = 1;

        for (int i = 1; i <= n; i++) {
            System.out.print(first + " ");

            int next = first + second;
            first = second;
            second = next;
        }

        sc.close();
    }
}
```

Q9. Print Factorial of a Number

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        long factorial = 1;

        for (int i = 1; i <= n; i++) {
            factorial = factorial * i;
        }

        System.out.println(factorial);

        sc.close();
    }
}
```

Q10. Remove Duplicates From Given String

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();
        String result = "";

        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);

            if (result.indexOf(ch) == -1) {
                result = result + ch;
            }
        }

        System.out.println(result);

        sc.close();
    }
}
```